

Unit 7, Lesson 2: Sectors of a Circle

Benchmark

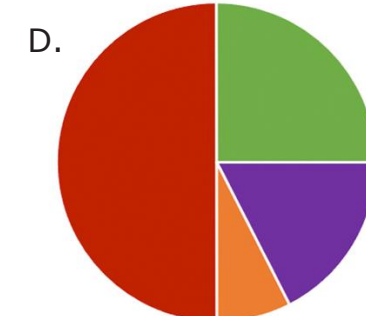
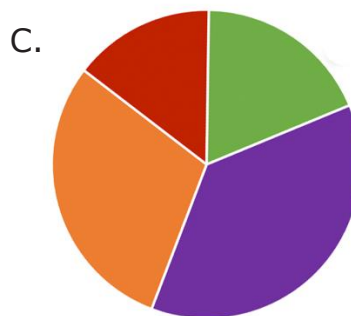
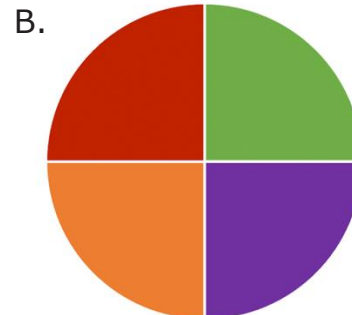
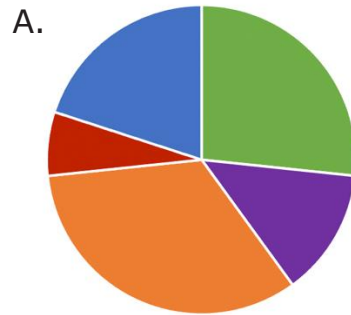
MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

Learning Targets

- ☐ I can find the percentage the area of a sector represents in a circle.
- ☐ I can find the central angle measure of a sector given the percentage of the circle a sector represents.

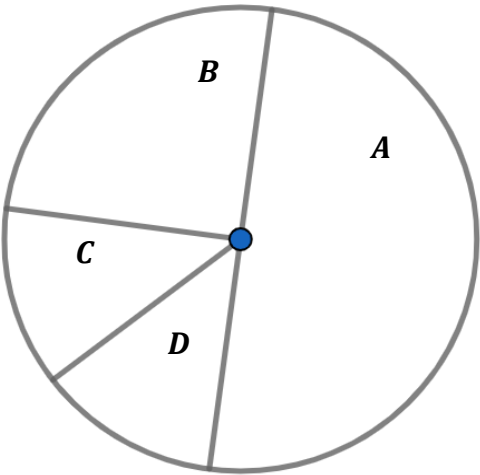
7.2.1 Warm-Up: Which One Doesn't Belong?

1. With your partner, review the four circle graphs and determine which one doesn't belong. Explain your choice.



7.2.2. Guided Instruction: Central Angle Measures of Sectors

1. The circle below is divided into sectors.

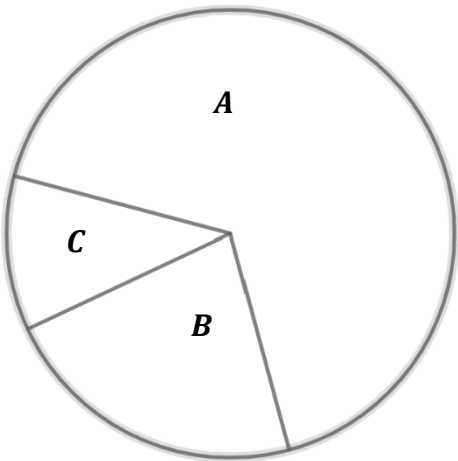


a. Use your protractor to measure the angle for each sector. Then, complete the table.

Sector	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
A			
B			
C			
D			

b. Discuss with your partner how the central angle is related to the percentage of the circle. Summarize your discussion.

2. A circle with three noncongruent sectors is shown.



a. Use your protractor to measure the angle for each sector. Then, complete the table. Round your answers to the nearest whole number.

Sector	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
<i>A</i>			
<i>B</i>			
<i>C</i>			
Total			

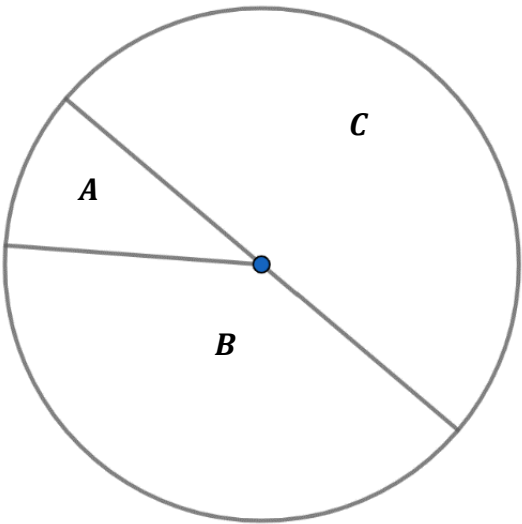
- b. What do you notice about the sums of the “total” row?
- c. Discuss with your partner why the total percentage may be slightly more or slightly less than 100%.

3. The table contains information for a circle with five different-sized sectors. Complete the table. Round your answers to the nearest whole number.

Sector	Percentage of the Circle	Ratio of Angle Measure to 360°	Central Angle Measure in Degrees
<i>A</i>	22%		
<i>B</i>	15%		
<i>C</i>	41%		
<i>D</i>	2%		
<i>E</i>			
Total	100%		

7.2.3 Guided Practice: Central Angle Measures

1. A circle with three noncongruent sectors is shown.



Use your protractor to measure the angle for each sector. Then, complete the table. Round your answers to the nearest whole number.

Fraction of the Circle	Central Angle Measure in Degrees	Ratio of Angle Measure to 360°	Percentage of the Circle
<i>A</i>			
<i>B</i>			
<i>C</i>			

2. The table contains information for a circle with four different size sectors. Complete the table. Round your answers to the nearest whole number.

Sector	Percentage of the Circle	Ratio of Angle Measure to 360°	Central Angle Measure in Degrees
<i>A</i>	36%		
<i>B</i>	24%		
<i>C</i>	17%		
<i>D</i>	23%		

7.2.4 Wrap-Up: Cool Down

1. Determine the measure of a central angle if the percentage of the circle is 63%. Round your answer to the nearest whole number.
2. What is the percentage of a circle if the central angle measures 40° ?